Lactate, Memory and Aging in Fruit Flies



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Introduction

The brain is a highly metabolically active tissue, thus requiring high levels of energy to be supplied consistently for proper cognitive function. There are several pathways for providing energy to neurons, the functional unit of the brain, however one in particular has drawn considerable recognition in recent years. This pathway involves the conversion of sugars into pyruvate and then from pyruvate to lactate (by the lactate dehydrogenase enzyme "LDH"), a metabolite which is believed to be passed from neighbouring, supporting cells towards the neurons when in immediate need of energy. As of late, studies have been suggesting that, in invertebrates such as *Drosophila melanogaster* (fruit fly), the production and release of lactate from supporting cells to the neurons promotes neuronal survival, synaptic function and axonal regeneration. Conversely, aberrant lactate metabolism can contribute to neurodegeneration. Although age-related issues are associated with cognitive deficits, previous studies have not yet determined whether lactate metabolism is correlated to age-related cognitive decline in *D. melanogaster*. To fill this knowledge gap, the Frame et al. (2023) [1] paper set out to explore whether changing lactate dehydrogenase levels in fruit flies will change their long-term memory as they age and to quantify this change. They tested their hypothesis by increasing (overexpression) and decreasing (knockdown) the enzyme (*dLdh*) levels in adult, male fruit fly neurons (thus creating three different genotypes, counting the control genotype). Subsequently, they measured the memory levels of the flies at 7, 14 and 21 days. They used fly courtship behaviour as a proxy for memory: flies that had previously been rejected by an already-mated female were considered trained flies, while flies that had not were considered naive flies; a trained fly that remembers the rejection courts less. As a reanalysis of their results, this project, in turn, explores the same question and checks whether the results obtained by the paper can be duplicated, using the same data, specifically the results showing how memory loss changes between different fly genotypes throughout their lifetime.

Methods

The data published by the Frame et al. (2023) paper was downloaded from Dryad [2], a public data repository. Since this project focused rather on the memory loss differences among fly genotypes than on the impact of lactate on life span or other metrics, most of the data obtained by the study was not used. Therefore, out of the 15 data files the paper published, this project only analysed 2 of the 7 sheets of one file (`Figure2B_2C.xlsx`). The two sheets contained the courtship indexes and memory indexes of the flies throughout the experiment. Further details on the organisation of data in the files can be found in the table in **Appendix A**. The data required no cleaning or prior processing besides the `dropna()` command which removed any empty cells from the files. Otherwise, there were no missing values in the files, no outliers were removed, nor units converted. The packages used in creating the plot and analysing the data were: `pandas` (for reading the files), `numpy` (for calculations such as bootstrap), `scipy.stats` (for the statistical tests) and `matplotlib` (for creating the plot).

The first step in analysing the data was the calculation of the memory index (MI) for every trained fly in each genotype and age, where 0 was the equivalent of no memory and 1 meant perfect memory. Next, the group mean and its uncertainties were calculated using the following formulas, where C_T is the trained CI, C_N the naive CI, s the standard deviation and n the number of flies:

$$MI_i = 1 - \frac{C_{T,i}}{\langle C_N \rangle}, \qquad \langle MI \rangle = 1 - \frac{\langle C_T \rangle}{\langle C_N \rangle}$$

$$\sigma_{\text{stat}} = \frac{s_T/\sqrt{n_T}}{\langle C_N \rangle}, \qquad \sigma_{\text{norm}} = \frac{\langle C_T \rangle}{\langle C_N \rangle^2} \frac{s_N}{\sqrt{n_N}}, \qquad \sigma = \sqrt{\sigma_{\text{stat}}^2 + \sigma_{\text{norm}}^2}$$

The statistical error is the standard error of the trained fly mean, while the second uncertainty is the error of the naive mean propagated through division. They are added in quadrature because they are uncorrelated (they come from different fly groups). Then, the control and mutant groups were compared by calculating the delta difference and its sigma with the following formula:

$$\Delta = \langle MI \rangle_{\text{control}} - \langle MI \rangle_{\text{mutant}}, \qquad \sigma_\Delta = \sqrt{\sigma_1^2 + \sigma_2^2}$$

Then, a bootstrap (10,000 resamples of the flies) was made to obtain the 95% range of the memory as a percentage of the control, since this is a ratio of two uncertain averages:

$$\% \text{ of control} = \frac{\langle MI \rangle_{\text{mutant}}}{\langle MI \rangle_{\text{control}}} \times 100$$

Lastly, two statistical tests were performed and an effect size was calculated. The Kruskal–Wallis test showed whether there was any difference between the three genotypes, Dunn's test (Bonferroni-corrected) was used to check the difference between the variant groups and the control, and Cliff's δ was used to quantify the differences found.

Results and discussion

Figure 1:

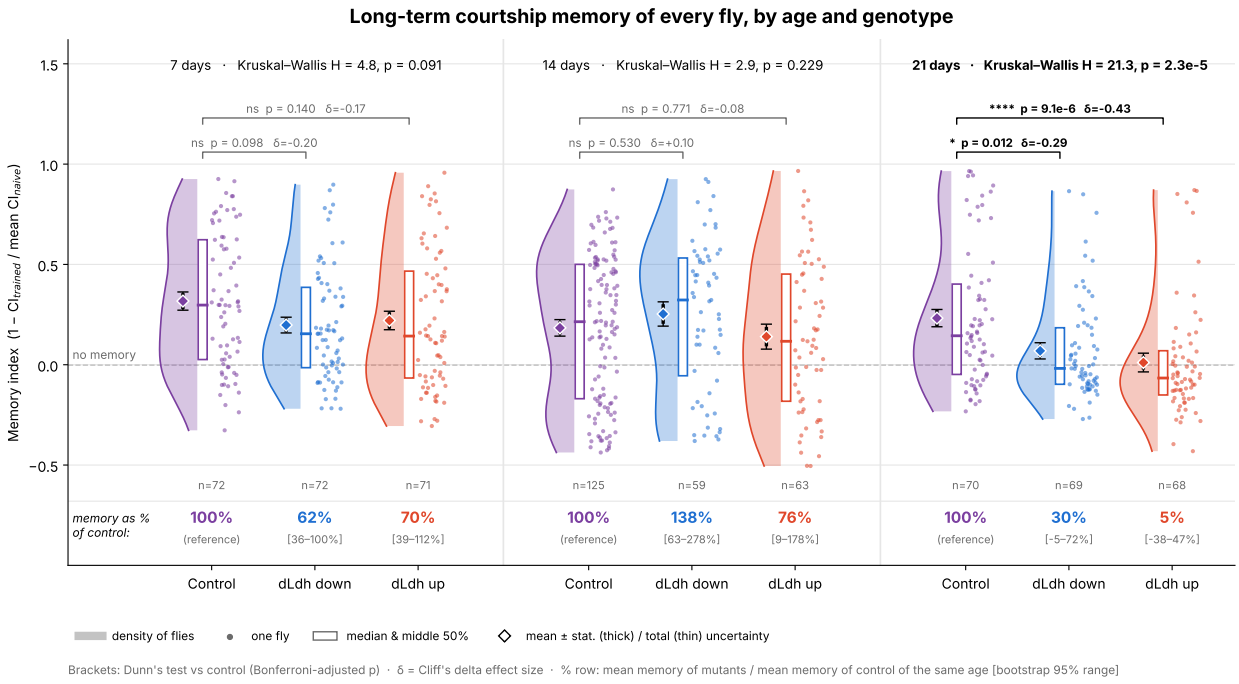


Figure 1: Long-term memory of individual flies at 7, 14 and 21 days for control (purple), *dLdh* knockdown (blue) and *dLdh* overexpression (red) in neurons. Dots: individual trained flies; half-violins: distribution; boxes: median and interquartile range; diamonds: mean ± statistical (thick) and total (thin) uncertainty. Dashed line: no memory. Top: Kruskal–Wallis test per age. Brackets: Dunn's test vs control (Bonferroni-adjusted p) with Cliff's δ. Bottom row: memory as % of control with bootstrap 95% range.

Based on the data analysis results, the plot shows that at 21 days, the mean memory index of flies with the enzyme overexpressed (0.011 ± 0.047) lies 3.5σ below that of control flies (0.232 ± 0.043). The statistical tests also confirm this difference, as Dunn's p-value for the comparison of the two groups is 9.1×10^{-6} and the effect size is medium to large ($\delta = -0.43$). Interestingly, as shown by the memory as % of control row below the plot, neither lowering nor increasing the enzyme levels improves memory, rather the contrary. These results imply that a balance in enzyme levels is ideal for memory retention throughout the flies' lifespans. This is particularly important as the plot shows no significant difference between the genotypes and control at young to medium stages of the flies' life (7 and 14 days). Importantly, the results of the data analysis performed for this project matched the values obtained by the Frame et al. (2023) paper, the Kruskal–Wallis p-value, more specifically.

Bibliography

[1] A. K. Frame et al., "Aging and memory are altered by genetically manipulating lactate dehydrogenase in the neurons or glia of flies," *Aging* 15(4), 947–981 (2023). doi:10.18632/aging.204565

[2] A. K. Frame et al., dataset on Dryad (2023). doi:10.5061/dryad.2v6wwpzsb

Appendix A: Organisation of the data file

The sheet `Courtship Indexes` of `Figure2B_2C.xlsx` contains 18 columns, one per age × genotype × training condition, named `<age> <genotype> <condition>` (e.g. `21d Control Trained`). Each cell is the courtship index of one fly. Rows have no meaning: naive and trained flies are different individuals, and columns have different lengths, so shorter columns end in empty cells. The other sheets contain the authors' own analysis and memory indexes, which were only used to check the results.

Table 1: Columns of the sheet `Courtship Indexes` and number of flies per column. Downregulation = neuronal *dLdh* knockdown; Upregulation = neuronal *dLdh* overexpression.

Genotype	Condition	7 days (`7d`)	14 days (`14d`)	21 days (`21d`)
Control	Naive	70	125	70
	Trained	72	125	70
Downregulation	Naive	71	63	65
	Trained	72	59	69
Upregulation	Naive	72	63	65
	Trained	71	63	68
Total: 1,333 flies		664 naive, 669 trained		